

CS5803 Data Visualisation

Superstore Sales Strategy Dashboard Report



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MSc Data Science and Analytics

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1. Introduction

1.1 Data

The dataset used in this project is the Sample Superstore Sales Dataset, publicly available on Kaggle (Vivek, 2022) at the following URL: [Superstore Dataset](#)

The dataset contains 9,994 rows and 21 columns representing transactional retail orders for a fictional US superstore from January 2014 to December 2017 across four regions and three product categories. The dataset contains zero missing values.

Data cleaning was performed in **R** including type verification, duplicate and missing value checks — confirming a clean dataset requiring no row removal. To enhance Tableau clarity and performance, non-essential and redundant columns such as RowID, Shipment date were removed, columns were converted from text strings to proper Date objects using `as.Date()` with format `'%m/%d/%Y'` and created four derived variables (profit margin %, profit status, discount band, and temporal variables) required for the visualization. The dataset of 9,994 rows and 17 columns was retained.

1.2 Data Dictionary

The following table lists all variables used in the final Tableau dashboard solution.

Variable	Type	Role	Domain/Range	Derived(Yes/No)
Order Date	Date	Time axis for SQ3	01/01/2014-31/12/2017	No- Converted from text to date
Segment	Dimension	CQ filter widget	Consumer, Corporate, Home office	No
State	Dimension	Geographic context	49 US states	No
Region	Dimension	SQ3 colour encoding, CQ heatmap column	East, West, Central, South	No
Category	Dimension	SQ1 Y axis , CQ brush source	Furniture, Office supplies, Technology	No
Sub-Category	Dimension	CQ heatmap row	17 Sub-Categories	No
Sales	Measure	SQ1 X axis, SQ3 Y axis	\$0.44 to \$22,638.48	No
Quantity	Measure	Supporting context	1 to 14 units per order	No
Discount	Measure	SQ2 X axis	0.0(0%) to 0.8(80%)	No
Profit	Measure	SQ2 Y axis, CQ heatmap colour	-\$6599.98 to +\$8399.98	No
profit_margin	Derived Measure	SQ1 colour encoding	-275 % to +50%	Yes- (Profit/Sales)*100
profit status	Derived Dimension	CQ tooltip label	Profitable or Loss Making	Yes- If profit>0,then

				profitable
discount band	Derived Dimension	SQ2 X axis grouping	5 Bands: From No Discount to Very High(31%)	Yes- binned from discount field
order_year	Derived Dimension	SQ3 time axis	2014, 2015, 2016,2017	Yes-extracted from order date
Order_quarter	Derived Dimension	SQ3 time axis granularity	Q1, Q2, Q3, Q4	Yes-extracted from order date

1.3 Persona and Questions

Jordan Blake is a Regional Sales Director at US Superstore, responsible for the commercial performance of all product categories across four regions. Jordan needs to understand which categories drive the most revenue, whether the company's discounting strategy is hurting profitability, how regional performance is trending overtime, and most critically which specific sub categories are generating losses that are being hidden by strong overall sales figures.

Simple Question 1 (SQ1) – Ranking

Which product categories generate the highest average sales per order and how does profit margin vary across these categories?

Relationship: Ranking

Justification: This question requires ranking three categories by two simultaneous measures-average sales and profit margin. Any single measure view or table cannot communicate the contrast in a single glance. A **bar chart** with colour encoding simultaneously reveals both dimensions, making the disconnect between sales performance and profitability immediately visible.

Variables: Category (Y axis), Sales-average (X axis), profit margin (colour encoding)

Simple question 2 (SQ2) – Impact

Does discounting help or hurt profitability and at what point does discounting start destroying value?

Relationship: Impact of Discount on Profit

Justification: The precise threshold where average profit turns negative cannot be detected from summary statistics and requires perceiving a pattern across grouped discount levels. A **binned bar chart** was used to represent this relationship.

Variables: discount band (X axis), Profit-average(Y axis), profit status (colour encoding)

Simple Question 3 (SQ3) – Time Series

How has total sales revenue evolved across the four regions between 2014 and 2017 and which regions are growing fastest?

Relationship: Time-Series and Comparison

Justification: Comparing four regional trend lines simultaneously on a shared time axis reveals crossing patterns, divergence, and seasonal peaks that are completely invisible in a table of numbers. A **multi-line chart** helps identify temporal trends, seasonal patterns, and differences in growth rates across regions.

Variables: Order.date, grouped by quarter(X axis), Sales-average(Y axis), Region (colour encoding)

Complex Question (CQ) – Multi-dimensional comparison

Among the highest selling categories, which specific sub-categories are quietly generating losses and are these losses concentrated in particular regions or universal?

Relationship: Multi-dimensional comparison

Justification: This question logically extends SQ1 by drilling down from category level to sub-category level, revealing which specific sub-categories are driving the low profitability. It involves four variables simultaneously — Category, Sub-Category, Profit and Region — which cannot be encoded together in a single readable static chart.

Interactions required: brushing from SQ1 to filter the heatmap; Segment and Year filters applying to CQ heatmap; **dynamically filtered heatmap** updating in response to SQ1 brush selection.

Variables: Category (brush source), Sub.Category (heatmap rows), Region (heatmap columns), Profit (cell colour), Segment (filter widget).

1.4 Requirements

R1 [SQ1 — Ranking]: A horizontal bar chart with Category on the Y axis and average Sales on the X axis, sorted in descending order, and each bar coloured by average profit_margin using a Red-Green diverging colour scale centered at 0%. This dual encoding enables Jordan to simultaneously perceive sales ranking from bar length and profitability from bar colour — a single chart communicating two measures.

R2 [SQ2 — Impact]: A binned bar chart with discount band on the X axis sorted from lowest to highest discount and avg-profit on the Y axis, bars coloured blue for profitable bands and red for loss-making bands, overlaid with a horizontal reference line at Profit = 0 labelled Break-even, would be appropriate..

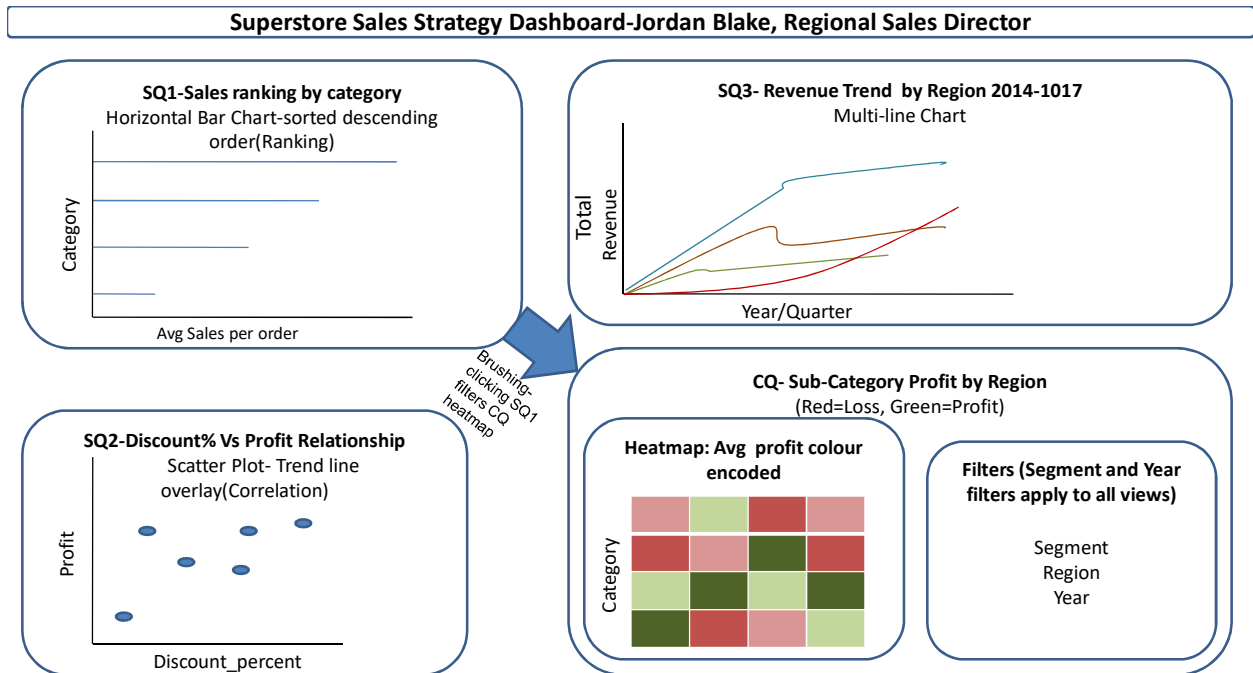
R3 [SQ3 — Time Series]: A multi-line chart with Order.Date grouped by quarter on the X axis, sum of Sales on the Y axis, and one distinctly coloured line per Region would be appropriate.

R4 [CQ — Multi-dimensional Comparison]: The CQ view is a heatmap with Sub.Category on the Y axis and Region on the X axis, each cell coloured by sum of Profit using a Red-Green diverging scale centred at zero. The SQ1 bar chart on the same dashboard is enabled as a filter — clicking a category bar directly filters the heatmap to show only that category's sub-categories (brushing interaction). A Segment dropdown filter and Year filter are also provided.

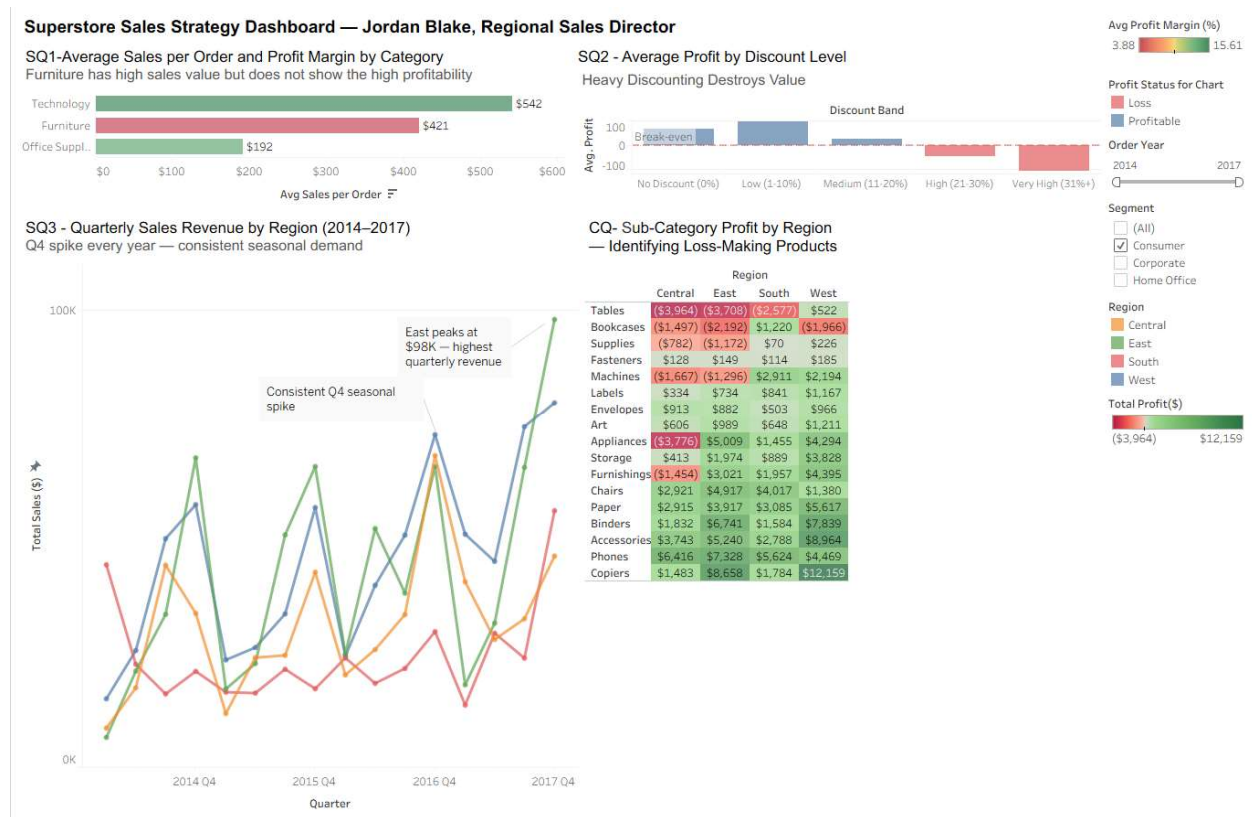
2. Design

2.1 Paper Landscape

The paper landscape design was created in PowerPoint before any Tableau implementation, showing a hypothetical four-view layout independent of Tableau's capabilities. It includes: a horizontal bar chart (SQ1, top left), a multi-line chart (SQ3, top right), a scatter plot (SQ2, bottom left), and a heatmap with filter panel (CQ, bottom right). A brushing arrow from SQ1 to the CQ heatmap is annotated, indicating the direct manipulation interaction along with the note that Segment and Year filters apply to all views simultaneously.



2.2 Annotated Dashboard Screenshot



Following Cairo (2013), simpler chart types were used for simple questions, with the heatmap reserved only where multi-dimensional data genuinely required it.

Change 1- SQ2: Scatter plot replaced with binned bar chart

The paper landscape proposed a scatter plot for SQ2. During implementation the Discount field proved to contain only twelve discrete values, causing thousands of data points to stack in twelve vertical columns making individual orders invisible and the correlation pattern unreadable. Jitter techniques were applied to spread the points but these distorted the data range — the X axis displayed impossible negative discount values — and still failed to communicate the relationship clearly. Additionally the negative correlation coefficient of -0.219 was too weak to produce a visually interpretable scatter pattern.

A binned bar chart was therefore adopted using the derived variable discount_band which groups orders into five meaningful discount levels. This chart type is more appropriate for data with discrete groupings, following Few's (2012) guidance that chart selection should be determined by what the data needs to communicate rather than prior design assumptions. The binned bar chart communicates the break-even threshold — the point at which average profit turns negative — far more clearly and directly than the scatter plot could. This change was judged an improvement over the original design rather than a compromise.

Change 2- Positioning of SQ2 and SQ3 charts

The paper landscape design placed SQ3 in the upper row alongside SQ1. During implementation SQ3 was moved to the lower row alongside the CQ heatmap, as this grouping better separates the two simple single-measure charts (SQ1 and SQ2) from the two multi-variable charts (SQ3 and CQ), creating a more logical visual hierarchy on the dashboard.

Change 3- Filters only apply to CQ Heatmap

In the paper landscape design, Segment and Year filters are shown in the filter panel, intended to apply to all views. In the final implementation these apply to the CQ heatmap only, as applying those to other charts distorted their visual encodings.

3. Implementation

The dashboard was implemented in Tableau Desktop 2023 using the cleaned dataset `superstore_clean.csv` containing 9,994 rows and 17 columns including four derived variables created in R.

3.1 SQ1- Horizontal Bar Chart

The SQ1 view was implemented by placing Category on the Rows shelf and the calculated field Avg Sales per Order — defined as $\text{SUM}(\text{Sales})/\text{COUNTD}(\text{Order.ID})$ — was placed on the Columns shelf, producing a horizontal bar chart sorted descending by sales value. The profit_margin derived variable was placed on the Color card using a Red-Green diverging scale with AVG aggregation, which is set as the default for this field. Visual encoding followed Mackinlay's (1986) effectiveness ranking — bar length encodes sales as position along a common scale is most accurately perceived, while colour saturation encodes profit margin where directional indication matters more than precise magnitude.

3.2 SQ2- Binned Bar Chart

The SQ2 view was implemented as a binned bar chart with discount_band variable on the Columns shelf as a discrete dimension, sorted logically from No Discount to Very High, and average Profit on the Rows shelf. Bars were coloured using a calculated field — blue for profitable bands, red for loss-making bands. A reference line at Profit = 0 was added via the Analytics panel, labelled Break-even with a dashed red line to make the profitability threshold immediately visible.

3.3 SQ3- Multi-Line Chart

The SQ3 view was implemented by placing Order.Date on the Columns shelf grouped by continuous Quarter and sum of Sales on the Rows shelf with the Y axis fixed to start at zero for honest quantitative comparison. Region was placed on the Color card producing four distinctly coloured trend lines. Data point markers were enabled on each trend line. The Y axis was formatted to start at zero following Few's (2012) guidance. Two annotations were added: one marking East region's peak quarterly revenue of \$98,023 in Q4 2017 and one highlighting the consistent Q4 seasonal spike visible every year.

3.4 CQ-Profit Heatmap

The CQ view was implemented as a heatmap by setting the Marks type to Square, placing Sub.Category on the Rows shelf and Region on the Columns shelf. Profit was placed on the Color card using a Red-Green diverging scale centered at zero — red cells indicate loss-making sub-category and region combinations, green cells indicate profitable combinations, and colour intensity encodes the magnitude of profit or loss. Sub-categories were sorted ascending by sum of Profit so Tables — the worst loss-maker at -\$17,725 — appears at the top making the most critical finding immediately visible.

Profit value labels formatted as currency with zero decimal places were added inside each cell via the Label card. A calculated field Profit Status Label was created and placed on the Detail card to enable meaningful tooltip text.

3.5 Dashboard Assembly and Interactivity

All four views were composed in a single 1366x900 fixed-size tiled dashboard. SQ1 and SQ2 sit side by side in Row 1 and SQ3 and CQ sit side by side in Row 2. The Use as Filter action was enabled on SQ1 — clicking any category bar dynamically filters both the CQ heatmap to show only that category's sub-categories and SQ2 to show the discount-profit relationship for that specific category, enabling coordinated multi-view exploration. Segment and Year filter widgets were added to the dashboard and apply to the CQ heatmap. The brushing and filtering interactions follow Spence's (2007) framework for direct manipulation in linked views. Tooltips were customised for each view to show contextually relevant information on hover. A Tableau Story sheet with four annotated story points was created to demonstrate each question answer in sequence.

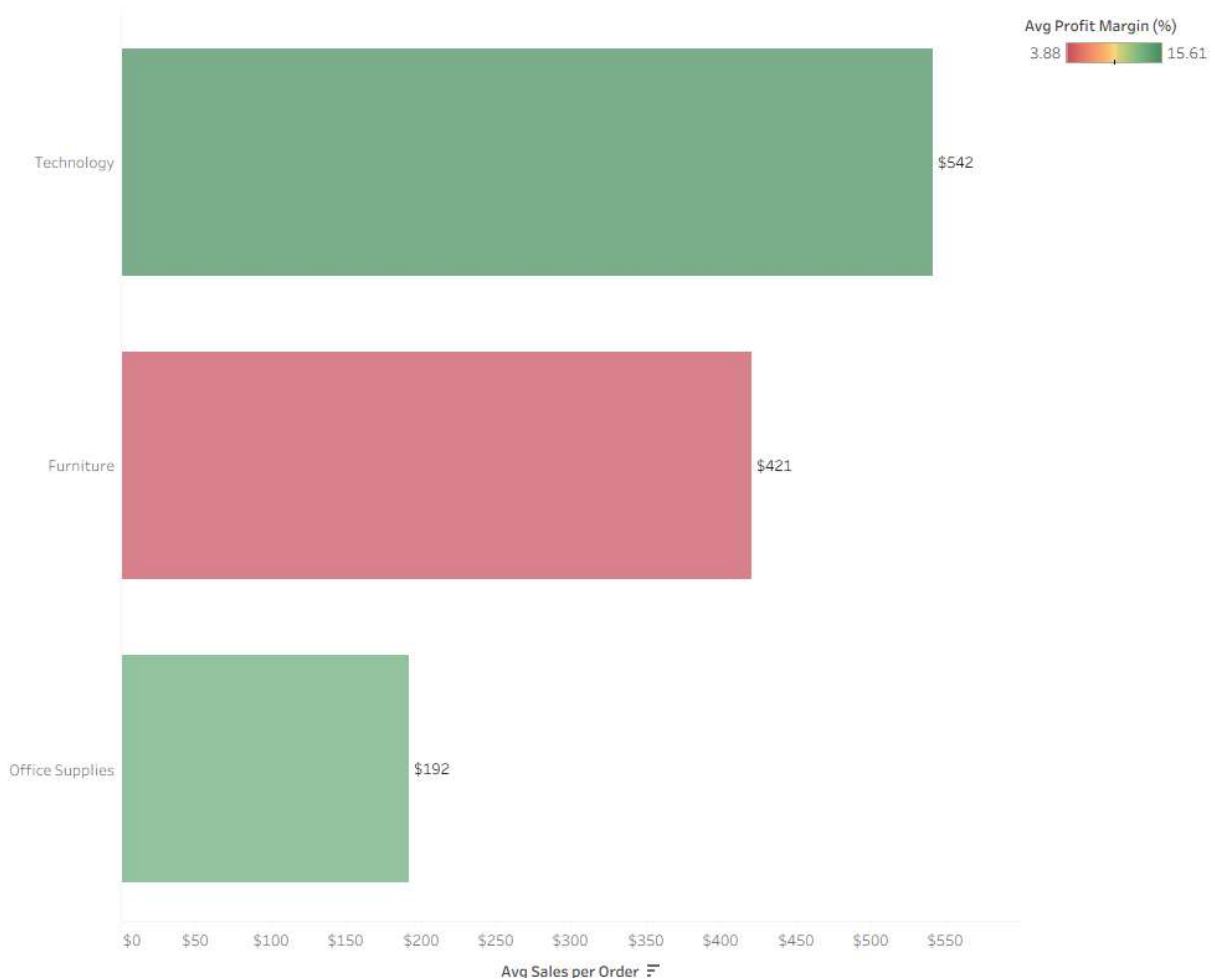
4. Walkthrough

SQ1- Sales ranking and Profitability Comparison (Story Point 1)

The horizontal bar chart immediately reveals a clear ranking of the three product categories. Technology leads with an average sale of \$542 per unique order, followed by Furniture at \$421 and Office Supplies at \$192. The colour encoding reveals Technology and Office Supplies bars are green (margins 15.61% and 13.80% respectively) while Furniture is red with a margin of only 3.88% — less than one quarter of Technology's margin despite being the second highest revenue category.

SQ1: Which categories generate the highest sales per order?

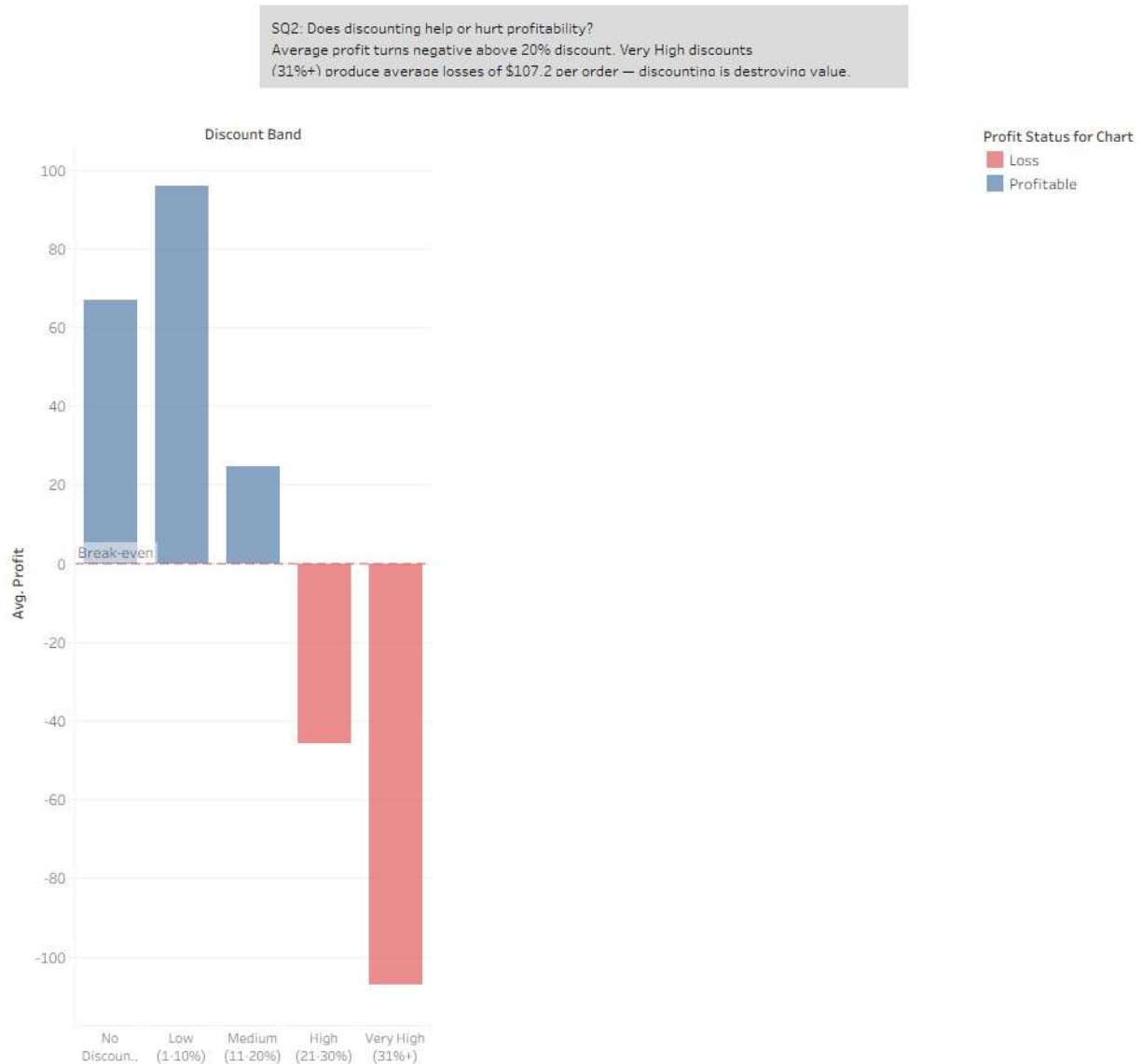
Technology leads sales at \$542 avg per order. Furniture earns \$421 but profit margin is only 3.88% vs Technology's 15.61%, indicates high sales categories do not always correspond to high profitability - highlighting inefficiencies in pricing or cost



SQ1 is answered: Technology is both the strongest revenue generator and most profitable. Furniture's strong sales conceal a serious profitability problem only visible through this dual encoding.

SQ2- Discount Level and Profitability(Story Point 2)

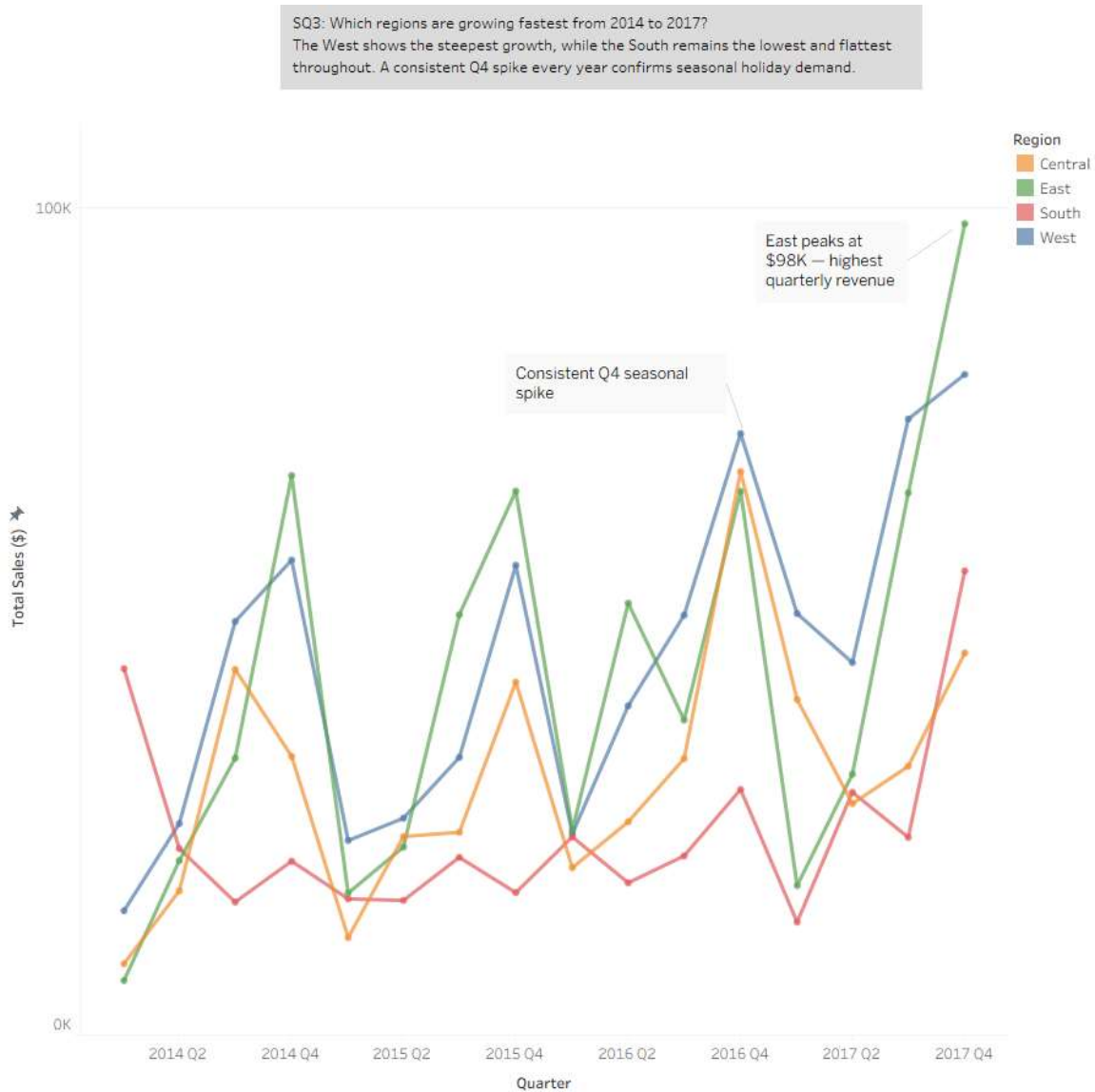
The binned bar chart shows average profit at each of the five discount bands. Average profit is \$66.90 at zero discount, rising to \$96.06 at low discounts of 1-10%. It falls to \$24.74 at medium discounts of 11-20%, then turns negative at high discounts of 21-30% (average loss \$45.68 per order) and deepens to -\$107.21 at very high discounts of 31%+. The break-even reference line makes the critical threshold immediately visible.



SQ2 is answered: Discounting above 20% actively destroys value.

SQ3- Regional Revenue Trends (Story Point 3)

The multi-line chart shows four regional revenue trend lines from Q1 2014 to Q4 2017. West region's line shows the steepest upward gradient confirming fastest growth. South's line remains flattest throughout. East's peak quarterly revenue of \$98K in Q4 2017 is annotated on the chart. A consistent Q4 spike is visible every year across all four lines confirming structural seasonal demand driven by holiday shopping.



SQ3 is answered: West grows fastest, South lags, and Q4 seasonality is universal.

CQ- Sub-Category Profitability by Region (Story Point 4)

The complex question required a two-step interactive investigation using the coordinated views.

Step 1: Clicking the Furniture bar in SQ1 triggered the brushing action, filtering the heatmap to show only Furniture sub-categories. Step 2: The heatmap immediately revealed Tables as entirely red across regions: Central -\$3,964, East -\$3,708, South -\$2,577, West +\$522. Tables generates a total loss of -\$17,725 — the worst-performing sub-category in the dataset. Bookcases also show losses in Central and East regions. Step 3: Applying the Segment filter to corporate customers — Tables losses persisted across all regions, confirming the problem is structural rather than customer-segment-driven.

Story Point 4 — CQ

CQ: Among top categories, which sub-categories are quietly losing money?
Selecting Furniture in the bar chart filters the heatmap — revealing Tables lose money in all four regions. This is a universal pricing crisis not a regional problem.



The CQ is answered: Tables losses are universal, not regional. This is a fundamental product pricing crisis. Jordan should consider repricing Tables significantly upward or removing them from the catalogue entirely. This finding was only discoverable through the coordinated brushing and filtering sequence.

5. Discussion

5.1 Critical Evaluation

The project successfully produced a dashboard that answers all four questions and produced a coherent business narrative: Technology leads sales and profit; heavy discounting destroys value above 20%; West is the fastest-growing region; and Tables are a universal loss-maker requiring urgent action. The strongest aspect of the implementation is the CQ heatmap which genuinely required interaction to answer and produced a non-obvious finding — that Tables losses are universal rather than regional. The weakest aspect was SQ2, as the original scatter plot was not feasible with the dataset structure. The substituted binned bar chart highlights the key threshold but sacrifices granular, order-level insight. This could be interpreted as an improvement driven by data structure constraints rather than a failure.

5.2 Strengths and weaknesses of Tableau

Tableau's drag-and-drop interface enabled rapid prototyping and the Use as Filter action provided accessible brushing interaction without programming. The diverging colour scale configuration was straightforward and produced clear profit/loss communication. However, several limitations emerged: the tiled layout system is imprecise, with restricted control over container sizing; default SUM aggregation can misrepresent percentage measures, requiring manual adjustment to AVG; and the lack of native scatter plot jitter prevents effective visualisation of discrete discount values, necessitating a binned bar chart (Section 2.2).

5.3 HCI and UCD Considerations

Nielsen's (1994) principle of progressive disclosure structures the layout, presenting summary insights upfront while reserving detailed analysis for interaction, thereby reducing cognitive load. Bertin's (1983) retinal variable theory and Mackinlay's (1986) effectiveness ranking guided visual encoding choices, prioritising position for quantitative comparison and colour for secondary variables and heatmap representation. Few's (2012) principles informed layout decisions, including allocating full width to time series and sorting the heatmap by loss-making sub-categories to prioritise actionable insights.

5.4 Personal Learning and Future Objectives

This project deepened understanding of the relationship between data structure and chart type appropriateness. The scatter plot failure demonstrated that a theoretically correct chart type can be practically unusable when the data lacks sufficient continuous variation. Learning to diagnose this and reason about alternatives using Few's (2012) framework was the most valuable practical skill developed. My future objective is to develop proficiency in **Power BI** as an alternative to Tableau, enabling direct comparison for business intelligence tasks. This would address the layout flexibility limitations encountered in this project while broadening my data visualisation toolkit in line with industry practice.

6. Conclusion

This project developed an interactive Tableau dashboard analysing the Superstore Sales Dataset (Vivek, 2022), comprising 9,994 US retail transactions from 2014–2017, to address three simple and one complex business question for Jordan Blake, Regional Sales Director. The analysis shows that Technology leads in both average sales (\$542 per order) and profitability (15.61% margin), while Furniture's strong sales mask a critically low 3.88% margin. Discounts above 20% consistently destroy value, with steep discounts generating average losses of \$107.21 per order. West region is the fastest growing region while South remains lowest throughout the entire period. Consistent Q4 peaks indicate structural seasonal demand. These findings are communicated through a horizontal bar chart, binned bar chart, multi-line chart, and interactive profit heatmap within a single fixed-size dashboard, incorporating brushing and filter interactions aligned with established data visualisation principles.

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AI Declaration: ChatGPT was used for R syntax guidance and to verify the correct Harvard referencing style in the report. Suggestions were reviewed and adapted selectely as a learning aid. All other work, including data selection, question formulations, design decisions, visualisation choices, written content, and critical reflections, is the author's own.

Appendix : All Supporting Project Files in a zip folder – 2550926_CS5803

1. 2550926_CS5803.dashboard.twbx
2. superstore_cleaned.csv
3. Superstore.R
4. Presentation1.pptx
5. 2550926_CS5803.docx